

CAA Test 2

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int main() {
    std::ofstream trace("cpu_trace.txt");
    int modeInput, cycles;
    std::cout << "Mode (0=DEFROST 1=HEAT 2=GRILL): ";
    std::cin >> modeInput;
    Mode mode = (Mode)modeInput;
    std::cout << "Number of cycles: ";
    std::cin >> cycles;

    State state = IDLE;
    uint8_t timer = 0;

    trace << "Cycle | State | Inputs | Outputs | Display\n";
    trace << std::string(65, '-') << "\n";

    for (int cycle = 1; cycle ≤ cycles; cycle++) {
        int s, st, d, f, t;
        std::cin >> s >> st >> d >> f >> t;
        bool START = s, STOP = st, DOOR = d, FOOD = f, TEMP = t;

        if (state ≠ ERROR_STATE) {
            if (state ≠ IDLE && state ≠ BUZZ) {
                if (TEMP) {
                    state = ERROR_STATE;
                } else if (!DOOR) {
                    state = PAUSE;
                } else if (STOP && state == RUN) {
                    state = PAUSE;
                }
            }
            switch (state) {
                case IDLE:
                    if (START && DOOR && FOOD) {
                        state = CHECK;
                        timer = modeCycles[mode];
                    }
                    break;
                case CHECK:
```

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        state = RUN;
        break;
    case RUN:
    {
        ALUResult res = alu_sub(timer, 1);
        timer = res.value;
        if (res.Z) state = COMPLETE;
    }
        break;
    case PAUSE:
        if (START && DOOR) {
            state = RUN;
        }
        break;
    case COMPLETE:
        state = BUZZ;
        break;
    case BUZZ:
        state = IDLE;
        timer = 0;
        break;
    case ERROR_STATE:
        break;
    }
}

bool heater    = (state == RUN);
bool turntable = (state == RUN);
bool light     = (state == CHECK || state == RUN || state ==
PAUSE || state == COMPLETE || state == ERROR_STATE);
bool buzzer    = (state == BUZZ || state == ERROR_STATE);

    logLine(trace, cycle, state, DOOR, FOOD, TEMP, heater,
turntable, light, buzzer, mode, timer);
}
trace.close();
std::cout << "\nTrace saved to cpu_trace.txt\n";
return 0;
}

```